

2.1 - Derivatives

Goals: Define the Derivative and apply it to initial examples

Before Class Videos: 2.1.1, 2.1.2

Example 1: Recall the velocity problem from the first day of class. If Trefor has some distance function $f(t)$, how fast is he going at 10 minutes? Let's explore graphically and numerically:

Definition:

The derivative of $f(x)$ at c is given by

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

provided this limit exists. If it does, we say f is differentiable at c .

Example 2: Compute the derivative of $f(x) = x^2$ at $x = 2$.

Definition:

The derivative of $f(x)$ on an open interval I is the function $f'(x)$ given by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

provided the limit exists for all x in I . If it does we say $f(x)$ is differentiable on I .

Example 3: $f(x) = 1/x^2$.

Example 4: $f(x) = \sqrt{x}$.

Example 5: Let's explore if the following functions are differentiable at $x = 0$:

a)

$$f(x) = \begin{cases} 2, & x \leq 0, \\ 1, & x > 0. \end{cases}$$

b) $f(x) = |x|$

c) $f(x) = x^{2/3}$

d) $f(x) = x^{1/3}$